

MODULE 9: ELASTIC BEANSTALK ASSIGNMENT

Problem Statement:

You work for XYZ Corporation. Your corporation wants to launch a new web-based application and they do not want their servers to be running all the time. It should also be managed by AWS. Implement suitable solutions.

Tasks To Be Performed:

1. Create an Elastic Beanstalk environment with the runtime as PHP.
2. Upload a simple PHP file to the environment once created.

SOLUTION:

Step 1: Create an Application

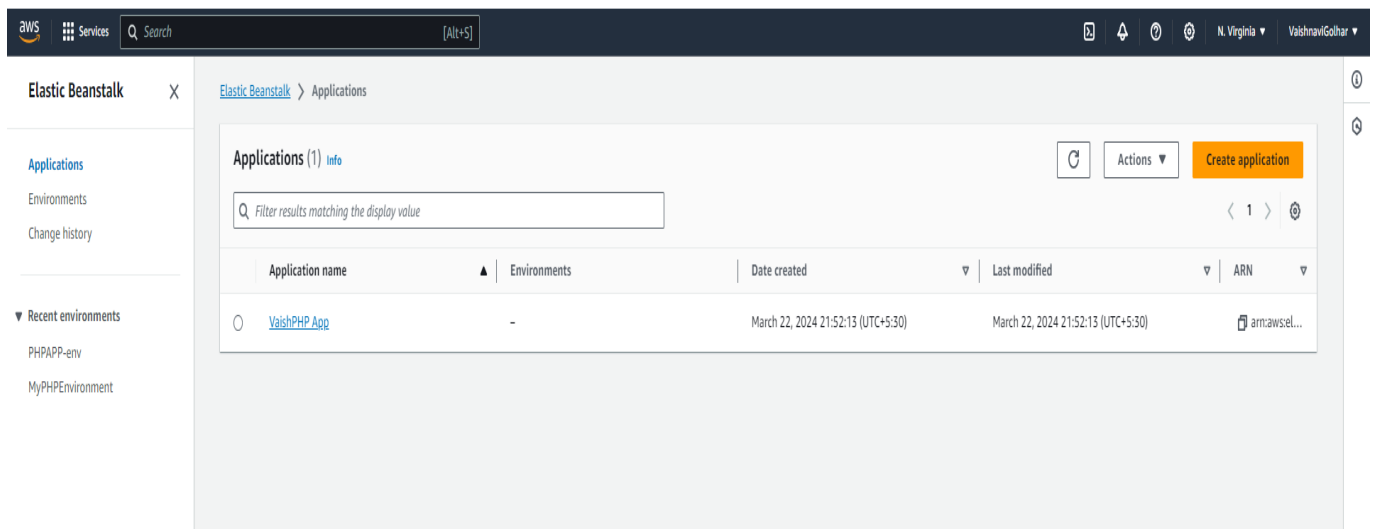
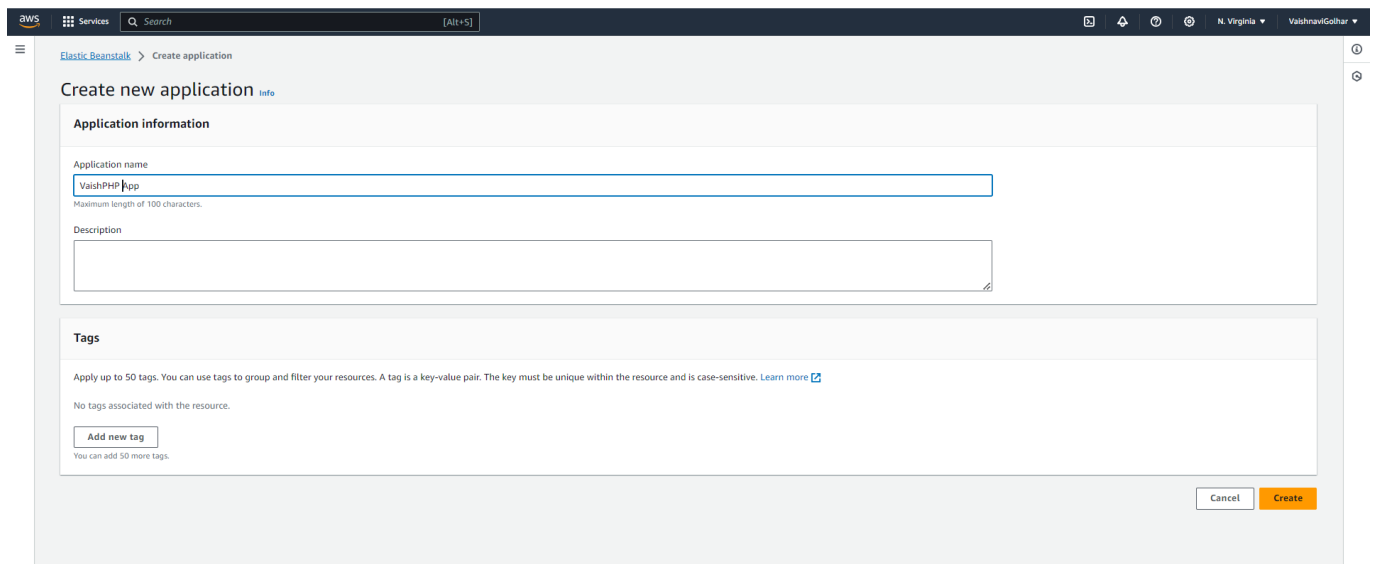
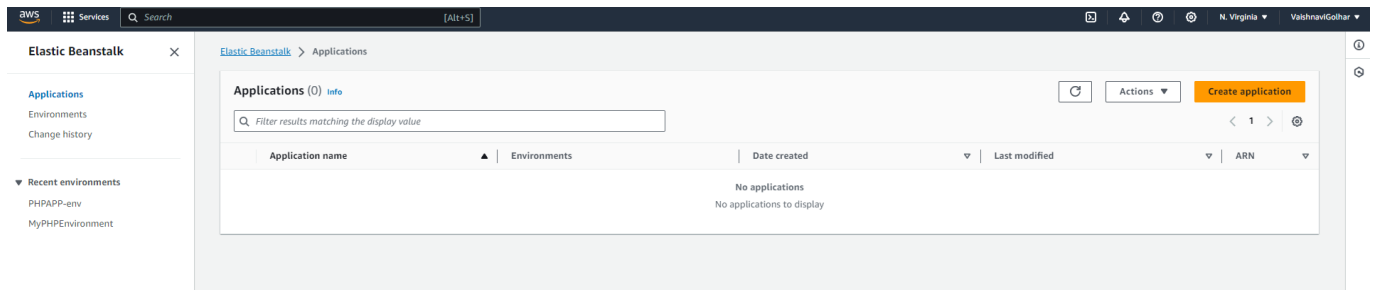
1. In the Elastic Beanstalk dashboard, click on the "Create Application" button.

Enter Application Details:

Provide a name for your application. This should be a descriptive name that represents your web-based application.

Optionally, you can provide a description to help identify the purpose of the application.

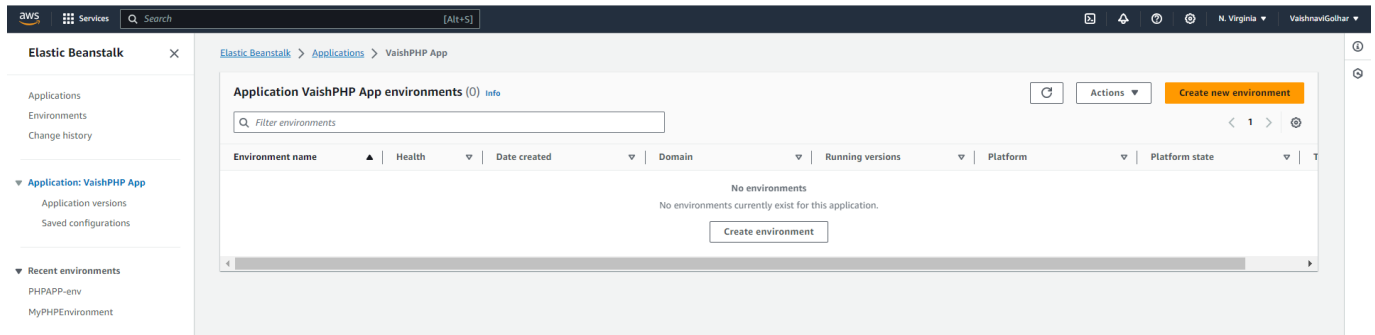
Click on "Create".



Step 2: Create an Environment

Select Application:

1. In the application dashboard, click on the "Create Environment" button.



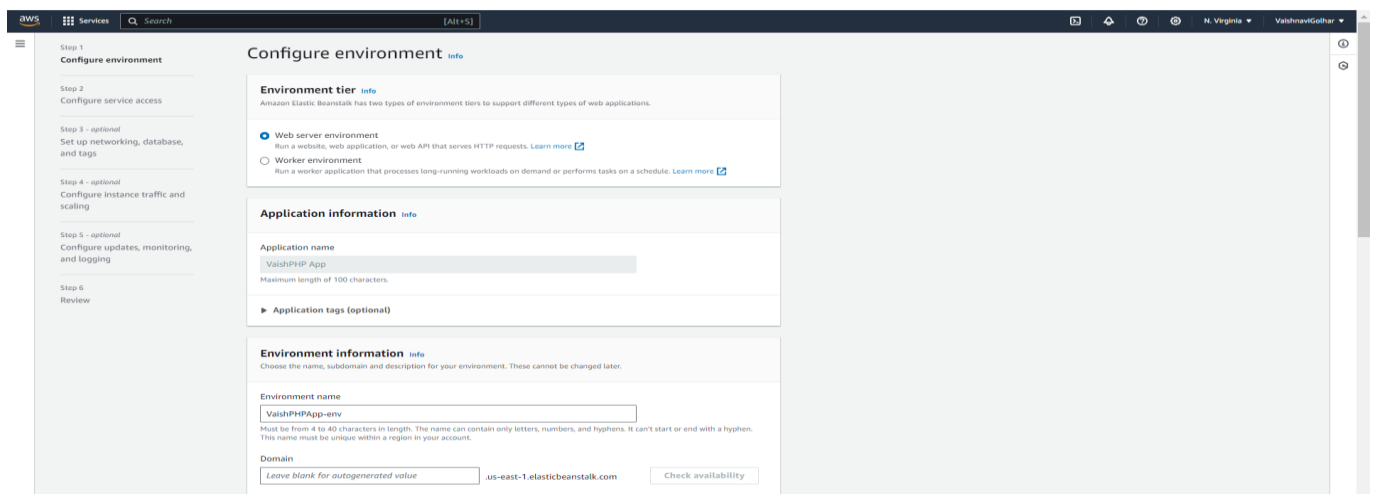
2. Choose Environment Tier:

Enter a name for your environment. This should be unique within your AWS account and should represent the environment's purpose.

Select the environment tier. For a basic web application, choose "Web server environment".

3. Choose Platform:

Select the platform for your application. Since you want to use PHP, choose "PHP" from the list of supported platforms.



The screenshot shows the AWS Elastic Beanstalk console. At the top, there's a search bar and navigation icons. The main content area is divided into two sections: 'Platform' and 'Application code'. The 'Platform' section has a 'Platform type' dropdown set to 'Managed platform', a 'Platform' dropdown set to 'PHP', a 'Platform branch' dropdown set to 'PHP 8.2 running on 64bit Amazon Linux 2023', and a 'Platform version' dropdown set to '4.1.0 (Recommended)'. The 'Application code' section has three radio buttons: 'Sample application', 'Existing version', and 'Upload your code' (which is selected). Below the radio buttons, there's a 'Version label' text input field and a 'Source code origin' section with a 'Local file' radio button selected. A 'Choose file' button is visible, and below it, a file named '1711119741169-index.php' is listed. At the bottom right, there are 'Cancel' and 'Next' buttons.

4.Configure Environment:

The screenshot shows the AWS Elastic Beanstalk console, specifically the 'Presets' configuration step. The 'Application code' section is still visible at the top, with 'Upload your code' selected. The 'Presets' section has a 'Configuration presets' dropdown set to 'Single instance (free tier eligible)'. Below this, there are four radio buttons: 'Single instance (free tier eligible)', 'Single instance (using spot instance)', 'High availability', and 'High availability (using spot and on-demand instances)'. At the bottom right, there are 'Cancel' and 'Next' buttons.

Configure other settings such as the domain, platform version, instance type, etc., according to your requirements.

aws

Services

Search

[Alt+S]

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Step 1
Configure environment

Step 2
Configure service access

Step 3 - optional
Set up networking, database, and tags

Step 4 - optional
Configure instance traffic and scaling

Step 5 - optional
Configure updates, monitoring, and logging

Step 6
Review

Configure service access

Service access

IAM roles, assumed by Elastic Beanstalk as a service role, and EC2 instance profiles allow Elastic Beanstalk to create and manage your environment. Both the IAM role and instance profile must be attached to IAM managed policies that contain the required permissions. [Learn more](#)

Service role

☒ Create and use new service role
☐ Use an existing service role

Service role name

Enter the name for an IAM role that Elastic Beanstalk will create to assume as a service role. Beanstalk will attach the required managed policies to it.

aws-elasticbeanstalk-service-role-01

[View permission details](#)

EC2 key pair

Select an EC2 key pair to securely log in to your EC2 instances. [Learn more](#)

FirstKey

[View permission details](#)

EC2 instance profile

Choose an IAM instance profile with managed policies that allow your EC2 instances to perform required operations.

[View permission details](#)

[Cancel](#) [Skip to review](#) [Previous](#) [Next](#)

aws

Services

Search

[Alt+S]

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Identity and Access Management (IAM)

Search IAM

IAM > Roles

Roles (14)

An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.

[Create role](#)

aws

Services

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IAM > Roles > Create role

Step 1
Select trusted entity

Step 2
Add permissions

Step 3
Name, review, and create

Select trusted entity

Trusted entity type

☒ **AWS service**
Allow AWS services like EC2, Lambda, or others to perform actions in this account.

☐ **AWS account**
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.

☐ **Web identity**
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.

☐ **SAML 2.0 federation**
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.

☐ **Custom trust policy**
Create a custom trust policy to enable others to perform actions in this account.

Use case

Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

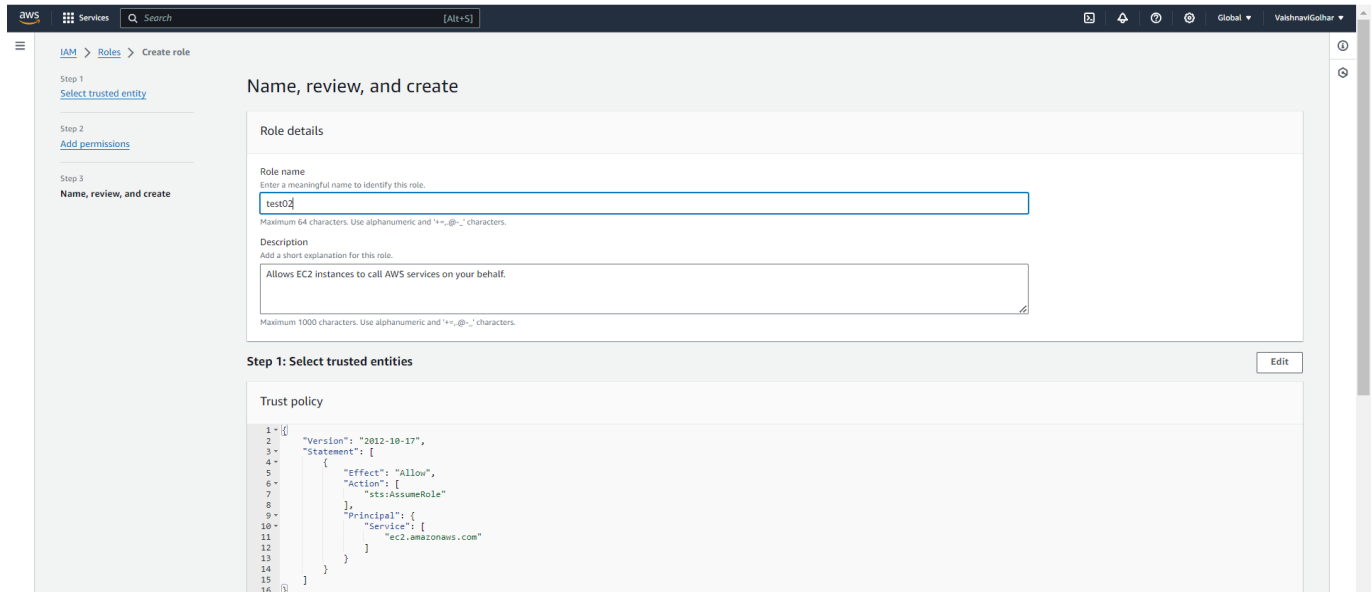
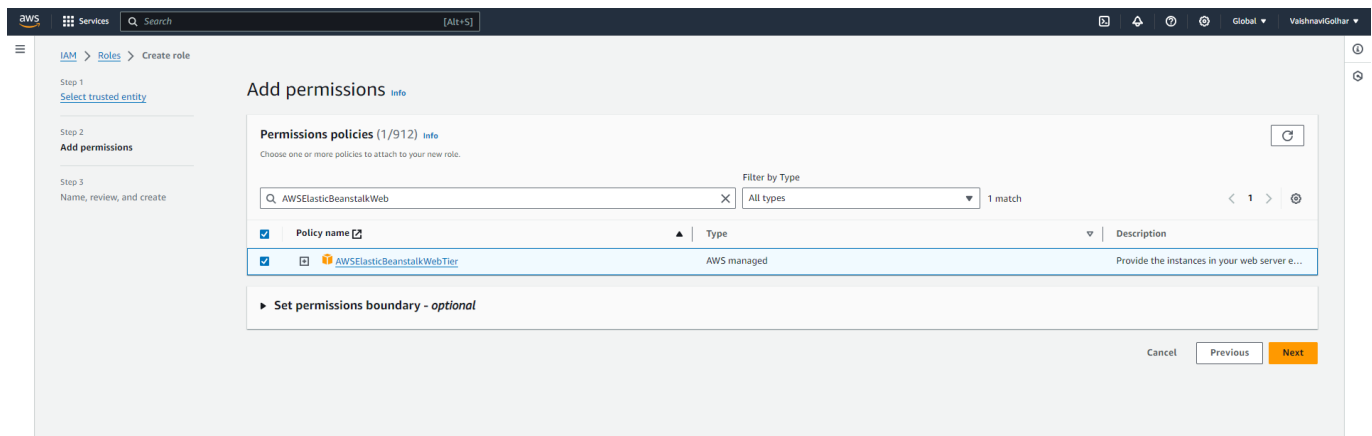
Service or use case

EC2

Choose a use case for the specified service.

Use case

☒ **EC2**
Allows EC2 instances to call AWS services on your behalf.



Identity and Access Management (IAM)

Search IAM

Dashboard

Access management

- User groups
- Users
- Roles**
- Policies
- Identity providers
- Account settings

Access reports

- Access Analyzer
- External access
- Unused access
- Analyzer settings
- Credential report
- Organization activity
- Service control policies (SCPs)

Related consoles

[IAM Identity Center](#)

[AWS Organizations](#)

IAM > Roles > test02

test02

info

Delete

Allows EC2 instances to call AWS services on your behalf.

Summary

Edit

Creation date

March 22, 2024, 22:01 (UTC+05:30)

ARN

arn:aws:iam::161681788148:role/test02

Instance profile ARN

arn:aws:iam::161681788148:instance-profile/test02

Last activity

-

Maximum session duration

1 hour

Permissions

Trust relationships

Tags

Access Advisor

Revoke sessions

Permissions policies (1)

info

Simulate

Remove

Add permissions

You can attach up to 10 managed policies.

Search

Filter by Type

All types

< 1 >

☐

Policy name

Type

Attached entities

☐

AWS Elastic Beanstalk WebTier

AWS managed

2

Permissions boundary (not set)

Generate policy based on CloudTrail events

Step 1

Configure environment

Step 2

Configure service access

Step 3 - optional

[Set up networking, database, and tags](#)

Step 4 - optional

[Configure instance traffic and scaling](#)

Step 5 - optional

[Configure updates, monitoring, and logging](#)

Step 6

[Review](#)

Configure service access

info

Service access

IAM roles, assumed by Elastic Beanstalk as a service role, and EC2 instance profiles allow Elastic Beanstalk to create and manage your environment. Both the IAM role and instance profile must be attached to IAM managed policies that contain the required permissions.

Learn more

Service role

Create and use new service role

Use an existing service role

Service role name

Enter the name for an IAM role that Elastic Beanstalk will create to assume as a service role. Beanstalk will attach the required managed policies to it.

aws-elasticbeanstalk-service-role-01

View permission details

EC2 key pair

Select an EC2 key pair to securely log in to your EC2 instances.

Learn more

FirstKey

EC2 instance profile

Choose an IAM instance profile with managed policies that allow your EC2 instances to perform required operations.

test02

View permission details

Cancel

Skip to review

Previous

Next

VPC dashboard

EC2 Global View

Filter by VPC:

Select a VPC

Virtual private cloud

Your VPCs

Subnets

Your VPCs (1)

info

Actions

Create VPC

Search

Name

VPC ID

State

IPv4 CIDR

IPv6 CIDR

DHCP option set

Main route table

Main network ACL

-

vpc-0b22f56ff603e7fa1

Available

172.31.0.0/16

-

dopt-0ab2b84ddcb3b6...

rtb-0b181079e58259c1f

acl-0da829dd447c...

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Services

Search

[Alt+S]

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Step 1
Configure environment

Step 2
Configure service access

Step 3 - optional
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Step 4 - optional
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Step 5 - optional
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Step 6
Review

Set up networking, database, and tags - optional

Virtual Private Cloud (VPC)

VPC
Launch your environment in a custom VPC instead of the default VPC. You can create a VPC and subnets in the VPC management console. [Learn more](#)

vpc-0b22f56ff603e7fa1 | (172.31.0.0/16)

[Create custom VPC](#)

Instance settings

Choose a subnet in each AZ for the instances that run your application. To avoid exposing your instances to the Internet, run your instances in private subnets and load balancer in public subnets. To run your load balancer and instances in the same public subnets, assign public IP addresses to the instances. [Learn more](#)

Public IP address
Assign a public IP address to the Amazon EC2 instances in your environment.
☐ Activated

Instance subnets

Filter instance subnets

Availability Zone	Subnet	CIDR	Name
☒ us-east-1d	subnet-02777af0...	172.31.32.0/20	
☐ us-east-1f	subnet-03f211264...	172.31.64.0/20	
☐ us-east-1e	subnet-04fb4ecb8...	172.31.48.0/20	
☐ us-east-1c	subnet-091853ce9...	172.31.16.0/20	

No tags associated with the resource.

Add new tag

You can add 50 more tags.

Cancel

Skip to review

Previous

Next

aws

Services

Search

[Alt+S]

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Step 1
Configure environment

Step 2
Configure service access

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Review

Configure instance traffic and scaling - optional

Instances

Configure the Amazon EC2 instances that run your application.

Root volume (boot device)

Root volume type
(Container default)

Size
The number of gigabytes of the root volume attached to each instance.
8 GB

IOPS
Input/output operations per second for a provisioned IOPS (SSD) volume.
100 IOPS

Throughput
The desired throughput to provision for the Amazon EBS root volume attached to your environment's EC2 instance
125 MiB/s

Amazon CloudWatch monitoring

The time interval between when metrics are reported from the EC2 instances

Monitoring interval
5 minute

Instance metadata service (IMDS)

Your environment's platform supports both IMDSv1 and IMDSv2. To enforce IMDSv2, deactivate IMDSv1. [Learn more](#)

IMDSv1

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Services

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VPC dashboard

EC2 Global View

Filter by VPC:

Select a VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

Endpoints

Endpoint services

NAT gateways

Peering connections

Security

Network ACLs

Security groups

Security Groups (1/3)

Find resources by attribute or tag

Actions

Export security groups to CSV

Create security group

	Name	Security group ID	Security group name	VPC ID	Description	Owner
☒	-	sg-0d5626455d6575a48	launch-wizard-1	vpc-0b22f56ff603e7fa1	launch-wizard-1 created 2024-03-01T...	161681788148
☐	-	sg-0c72b12d000b3d025	ApplicationSecurityGroup	vpc-0b22f56ff603e7fa1	allow ssh from public subnet of web s...	161681788148
☐	WebSecurityGroup	sg-08693198bc1703d9e	default	vpc-0b22f56ff603e7fa1	default VPC security group	161681788148

sg-0d5626455d6575a48 - launch-wizard-1

DetailsInbound rulesOutbound rulesTags

Inbound rules (1)

Manage tagsEdit inbound rules

Search

	Name	Security group rule...	IP version	Type	Protocol	Port range	Source	Description
☐	-	sgr-0c06b256934a671...	IPv4	All traffic	All	All	0.0.0.0/0	-

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Monitoring interval

5 minute

Instance metadata service (IMDS)

Your environment's platform supports both IMDSv1 and IMDSv2. To enforce IMDSv2, deactivate IMDSv1. Learn more

IMDSv1

With the current setting, the environment enables only IMDSv2.

☒ Deactivated

EC2 security groups

Select security groups to control traffic.

EC2 security groups (3)

Filter security groups

	Group name	Group ID	Name
☐	ApplicationSecurityGroup	sg-0c72b12d000b3d025	
☐	default	sg-08693198bc1703d9e	
☒	launch-wizard-1	sg-0d5626455d6575a48	

Capacity

Configure the compute capacity of your environment and auto scaling settings to optimize the number of instances used.

Auto scaling group

Environment type

Select a single-instance or load-balanced environment. You can develop and test an application in a single-instance environment to save costs and then upgrade to a load-balanced environment when the application is ready for production. Learn more

Single instance

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Services

Search

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Capacity Rebalancing

Specifies whether to enable the capacity rebalancing feature for Spot Instances in your Auto Scaling Group. This option is only relevant when EnableSpot is true in the `aws-ec2` instances namespace, and there is at least one Spot Instance in your Auto Scaling group.

☐ Turn on capacity rebalancing

Architecture

The processor architecture determines the instance types that are made available. You can't change this selection after you create the environment. [Learn more](#)

☒ x86_64
This architecture uses x86 processors and is compatible with most third-party tools and libraries.

☐ arm64 - new
This architecture uses AWS Graviton2 processors. You might have to recompile some third-party tools and libraries.

Instance types

Add instance types for your fleet. Change the order that the instances are in to set the preferred launch order. This only affects On-Demand instances. We recommend you include at least two instance types. [Learn more](#)

Choose x86 instance types

t3.micro x t3.small x

AMI ID

Elastic Beanstalk selects a default Amazon Machine Image (AMI) for your environment based on the Region, platform version, and processor architecture that you choose. [Learn more](#)

ami-037c9b6cd5a6bac1

Availability Zones

Number of Availability Zones (AZs) to use.

Any

Placement

Specify Availability Zones (AZs) to use.

Choose Availability Zones (AZs)

Scaling cooldown

360 seconds

Cancel

Skip to review

Previous

Next

aws

Services

Search

[Alt+S]

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Step 1

Configure environment

Step 2

Configure service access

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Step 6

Review

Configure updates, monitoring, and logging - optional

Monitoring

Health reporting

Enhanced health reporting provides free real-time application and operating system monitoring of the instances and other resources in your environment. The `EnvironmentHealth` custom metric is provided free with enhanced health reporting. Additional charges apply for each custom metric. For more information, see [Amazon CloudWatch Pricing](#)

System

☐ Basic

☒ Enhanced

CloudWatch Custom Metrics - Instance

Choose metrics

CloudWatch Custom Metrics - Environment

Choose metrics

Health event streaming to CloudWatch Logs

Configure Elastic Beanstalk to stream environment health events to CloudWatch Logs. You can set the retention up to a maximum of ten years and configure Elastic Beanstalk to delete the logs when you terminate your environment.

Log streaming

☐ Activated (standard CloudWatch charges apply.)

Retention

7

Lifecycle

Keep logs after terminating environment

The screenshot shows the AWS Elastic Beanstalk configuration console. The interface is divided into several sections for configuring the environment:

- General Settings:**
 - Activated: 100 (Percentage)
 - Command timeout: 600
 - Deployment policy: AllAtOnce
 - Health threshold: Ok
 - Ignore health check: false
 - Instance replacement: false
- Platform software:**
 - Lifecycle: false
 - Log streaming: Deactivated
 - Allow URL fopen: On
 - Display errors: Off
 - Document root: -
 - Max execution time: 60
 - Memory limit: 256M
 - Zlib output compression: Off
 - Proxy server: nginx
 - Logs retention: 7
 - Rotate logs: Deactivated
 - Update level: minor
 - X-Ray enabled: Deactivated
- Environment properties:**
 - A table with columns 'Key' and 'Value'.
 - Message: "No environment properties. There are no environment properties defined."

At the bottom, there are three buttons: "Cancel", "Previous", and "Submit".

Click on "Create Environment".

5.Wait for Environment Creation:

Elastic Beanstalk will now create the environment based on your configurations. This may take a few minutes.

Environment Details:

Once the environment is created, you'll be redirected to the environment dashboard.

Here, you can view details such as the environment URL, health status, configuration, etc.

Services

Search

[Alt+S]

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Elastic Beanstalk

Applications

Environments

Change history

▼ Application: VaishPHPApp

Application versions

Saved configurations

▼ Environment: VaishPHPApp-env

Go to environment

Configuration

Events

Health

Logs

Monitoring

Alarms

Managed updates

Tags

▼ Recent environments

VaishPHPApp-env

MyPHPEnvironment

Successfully uploaded file php.zip to S3, created application version and started deployment with new application version

Elastic Beanstalk is updating your environment. To cancel this operation select **Abort Current Operation** from the Actions dropdown.

March 22, 2024 22:21:33 (UTC+5:30)	INFO	Deploying new version to instance(s).
March 22, 2024 22:21:14 (UTC+5:30)	INFO	Updating environment VaishPHPApp-env's configuration settings.
March 22, 2024 22:21:07 (UTC+5:30)	INFO	Environment update is starting.
March 22, 2024 22:19:27 (UTC+5:30)	INFO	Removed instance [i-0e5ede6ba89064c5e] from your environment.
March 22, 2024 22:18:27 (UTC+5:30)	INFO	Added instance [i-0cd12cd1d0c6b664a] to your environment.
March 22, 2024 22:18:09 (UTC+5:30)	INFO	Environment update completed successfully.
March 22, 2024 22:18:09 (UTC+5:30)	INFO	Successfully deployed new configuration to environment.
March 22, 2024 22:17:17 (UTC+5:30)	INFO	Created Auto Scaling launch configuration named: awseb-e-z6hptvw8ad-stack-AWSEBAutoScalingLaunchConfiguration-X85VMxSucXzn
March 22, 2024 22:17:01 (UTC+5:30)	INFO	Updating environment VaishPHPApp-env's configuration settings.
March 22, 2024 22:16:55 (UTC+5:30)	INFO	Environment update is starting.
March 22, 2024 22:14:52 (UTC+5:30)	ERROR	Create environment operation is complete, but with errors. For more information, see troubleshooting documentation.
March 22, 2024 22:14:27 (UTC+5:30)	INFO	Added instance [i-0e5ede6ba89064c5e] to your environment.
March 22, 2024 22:13:47 (UTC+5:30)	INFO	Command execution completed on all instances. Summary: [Successful: 0, Failed: 1].
March 22, 2024 22:13:47 (UTC+5:30)	ERROR	[Instance: i-0e5ede6ba89064c5e] Command failed on instance. Return code: 1 Output: Engine execution has encountered an error..
March 22, 2024 22:13:44 (UTC+5:30)	ERROR	Instance deployment failed. For details, see 'eb-engine.log'.
March 22, 2024 22:13:44 (UTC+5:30)	ERROR	Instance deployment: Your source bundle has issues that caused the deployment to fail. For details, see 'eb-engine.log'.
March 22, 2024 22:13:10 (UTC+5:30)	INFO	Waiting for EC2 instances to launch. This may take a few minutes.
March 22, 2024 22:12:50 (UTC+5:30)	INFO	Created EIP: 54.82.199.212

Upload and deploy

✕

📘 To deploy a previous version, go to the [Application versions page](#)

Upload application

📁 Choose file

🟢 File name: **php.zip**

File must be less than 500MB max file size

Version label

Unique name for this version of your application code.

1-3

Current number of EC2 instances: **1**

Cancel

Deploy

aws

Services

Search

[Alt+S]

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EC2 Dashboard

EC2 Global View

Events

Console-to-Code

Instances

Instance Types

Launch Templates

Spot Requests

Instances (1) info

Find Instance by attribute or tag (case-sensitive)

All states

Instance state = running

Clear filters

Name

Instance ID

Instance state

Instance type

Status check

Alarm status

Availability Zone

Public IPv4 DNS

Public IPv4 ...

Elastic IP

IPv6 IPs

VaishPHPApp-...

i-0e5ede6ba89064c5e

Running

t3.micro

Initializing

View alarms

us-east-1d

ec2-54-82-199-212.co...

54.82.199.212

54.82.199.212

-

Congratulations!

Your AWS Elastic Beanstalk *PHP* application is now running on your own dedicated environment in the AWS Cloud

You are running PHP version 8.2.15

This environment is launched with Elastic Beanstalk PHP Platform

What's Next?

- [AWS Elastic Beanstalk overview](#)
- [Deploying AWS Elastic Beanstalk Applications in PHP Using Eb and Git](#)
- [Using Amazon RDS with PHP](#)
- [Customizing the Software on EC2 Instances](#)
- [Customizing Environment Resources](#)

AWS SDK for PHP

- [AWS SDK for PHP home](#)
- [PHP developer center](#)
- [AWS SDK for PHP on GitHub](#)